

In[*]:=

```
<< mGRG`STensor`
```

Chap. 2 An Introduction to Tensors and Differential Forms

2.1 Tensors of Rank 1: Vectors

2.1.1 Covariant Vectors

Covariant vector: v_a

2.1.2 Contravariant Vectors

Contravariant vector: v^a

2.2 Tensors of Higher Rank

Rank-3 tensor: t_a^{bc}

2.3 Exterior Differential Forms

Differential: dx^a

Exterior or Wedge product: $dx^a \wedge dx^b = -dx^b \wedge dx^a$

1-form: $\omega = \omega_a dx^a$

2-form: $\omega = \frac{1}{2!} \omega_{[ab]} dx^a \wedge dx^b$

3-form: $\omega = \frac{1}{3!} \omega_{[abc]} dx^a \wedge dx^b \wedge dx^c$

2.4 The Lie Derivative

Lie derivative of T_a^b along the v -direction:

$$\mathcal{L}_v T_a^b = v^p \partial_p T_a^b + (\partial_a v^p) T_p^b - (\partial_p v^b) T_a^p$$

2.5 Affine Connection, Covariant Derivative, Torsion

Covariant derivative:

$$\nabla_a T_b^c = \partial_a T_b^c - \Gamma_{ab}^p T_p^c + \Gamma_{ap}^c T_b^p$$

Torsion in coordinate basis:

$$t_{ab}^c = \Gamma_{ab}^c - \Gamma_{ba}^c$$

벡터 v 와 텐서 T 를 정의한다:

In[*]:=

```
Tdefine[T, 2]; Tdefine[v, 1]
```

벡터 v 방향으로 텐서 T 의 Lie 도함수:

```
In[*]:= LD[v, T[ua, ub]]
```

```
Out[*]=
```

$$\mathcal{L}_v T^{ab}$$

Lie 도함수를 torsion-free 공변 도함수로 바꾼다:

$$\mathcal{L}_v T^{ab} = v^p \nabla_p T^{ab} - (\nabla_p v^a) T^{pb} - (\nabla_p v^b) T^{ap}$$

```
In[*]:= % // LDtoCD
```

```
Out[*]=
```

$$-\nabla_c v^b T^{ac} - \nabla_c v^a T^{cb} + \nabla_c T^{ab} v^c$$

공변 도함수를 보통 도함수로 바꾼 후, 단순화한다:

```
In[*]:= % // CDtoBD
```

```
% // Tsimplify
```

```
Out[*]=
```

$$-\partial_c v^b T^{ac} - \partial_c v^a T^{cb} + \partial_c T^{ab} v^c + \Gamma_{cd}^b T^{ad} v^c + \Gamma_{cd}^a T^{db} v^c - \Gamma_{cd}^b T^{ac} v^d - \Gamma_{cd}^a T^{cb} v^d$$

```
Out[*]=
```

$$-\partial_c v^b T^{ac} - \partial_c v^a T^{cb} + \partial_c T^{ab} v^c$$

유사한 방법으로 Lie 도함수를 보통 도함수로 바꾼다:

$$\mathcal{L}_v T_{ab} = v^p \nabla_p T_{ab} + (\nabla_a v^p) T_{pb} + (\nabla_b v^p) T_{ap}$$

```
In[*]:= LD[v, T[la, lb]]
```

```
% // LDtoCD
```

```
% // CDtoBD
```

```
% // Tsimplify
```

```
Out[*]=
```

$$\mathcal{L}_v T_{ab}$$

```
Out[*]=
```

$$\nabla_b v^c T_{ac} + \nabla_a v^c T_{cb} + \nabla_c T_{ab} v^c$$

```
Out[*]=
```

$$\partial_b v^c T_{ac} + \partial_a v^c T_{cb} + \partial_c T_{ab} v^c + \Gamma_{bc}^d T_{ad} v^c - \Gamma_{cb}^d T_{ad} v^c + \Gamma_{ac}^d T_{db} v^c - \Gamma_{ca}^d T_{db} v^c$$

```
Out[*]=
```

$$\partial_b v^c T_{ac} + \partial_a v^c T_{cb} + \partial_c T_{ab} v^c$$

$$\mathcal{L}_v T^a_b = v^p \nabla_p T^a_b - (\nabla_p v^a) T^p_b + (\nabla_b v^p) T^a_p$$

```
In[*]:= LD[v, T[ua, lb]]
% // LDtoCD
% // CDtoBD
% // Tsimplify
```

```
Out[*]=
 $\mathcal{L}_v T^a_b$ 
```

```
Out[*]=
 $\nabla_b v^c T^a_c - \nabla_c v^a T^c_b + \nabla_c T^a_b v^c$ 
```

```
Out[*]=
 $\partial_b v^c T^a_c - \partial_c v^a T^c_b + \partial_c T^a_b v^c + \Gamma_{bc}^d T^a_d v^c - \Gamma_{cb}^d T^a_d v^c + \Gamma_{cd}^a T^d_b v^c - \Gamma_{cd}^a T^c_b v^d$ 
```

```
Out[*]=
 $\partial_b v^c T^a_c - \partial_c v^a T^c_b + \partial_c T^a_b v^c$ 
```

2.6 The Metric Tensor

Line element:

$$ds^2 = g_{ab} dx^a dx^b$$

2.6.1 Affine Connection and the Metric

In the torsion-free spacetime and in the coordinate basis, the covariantly constant condition of g_{ab} , $\nabla_a g_{bc} = 0$, gives the Christoffel symbols

$$\Gamma_{ab}^c = \frac{1}{2} g^{cd} (\partial_a g_{bd} + \partial_b g_{ad} - \partial_d g_{ab})$$

2.6.2 Geodesics

$$v^a = \frac{dx^a}{d\lambda}, \quad v^a \nabla_a v^b = 0$$

```
In[*]:= v[ua] x CD[la, v[ub]]
% // CDtoBD
```

```
Out[*]=
 $\nabla_a v^b v^a$ 
```

```
Out[*]=
 $\partial_a v^b v^a + \Gamma_{ac}^b v^a v^c$ 
```

2.6.3 Lowering and Raising indices

$$v^a = g^{ab} v_b, \quad v_a = g_{ab} v^b$$

2.7 The Riemann Curvature Tensor

Torsion-free 곡률 텐서의 정의:

$$(\nabla_a \nabla_b - \nabla_b \nabla_a) v^d = -R_{abc}^d v^c$$

```

In[*]:= Tdefine[v, 1]

In[*]:= CD[la, lb, v[ud]] - CD[lb, la, v[ud]] (*torsion-free*)
% // CDtoBD
% // CDtoBD
% // Tsimplify

Out[*]=

$$\nabla_a \nabla_b v^d - \nabla_b \nabla_a v^d$$


Out[*]=

$$\partial_a \nabla_b v^d - \partial_b \nabla_a v^d - \nabla_c v^d \Gamma_{ab}^c + \nabla_b v^c \Gamma_{ac}^d + \nabla_c v^d \Gamma_{ba}^c - \nabla_a v^c \Gamma_{bc}^d$$


Out[*]=

$$\begin{aligned} & \partial_a \partial_b v^d - \partial_b \partial_a v^d - \partial_c v^d \Gamma_{ab}^c + \partial_c v^d \Gamma_{ba}^c + \partial_a \Gamma_{bc}^d v^c - \\ & \partial_b \Gamma_{ac}^d v^c - \Gamma_{ac}^e \Gamma_{be}^d v^c + \Gamma_{ac}^d \Gamma_{be}^c v^e - \Gamma_{ab}^c \Gamma_{ce}^d v^e + \Gamma_{ba}^c \Gamma_{ce}^d v^e \end{aligned}$$


Out[*]=

$$\partial_a \Gamma_{bc}^d v^c - \partial_b \Gamma_{ac}^d v^c - \Gamma_{ac}^e \Gamma_{be}^d v^c + \Gamma_{ac}^d \Gamma_{be}^c v^e$$


In[*]:= -RiemannCD[la, lb, lc, ud] × v[uc]

Out[*]=

$$-R_{abc}^d v^c$$


```

따라서

$$R_{abc}^d = -[\partial_a \Gamma_{bc}^d + \Gamma_{ap}^d \Gamma_{bc}^p - (a \leftrightarrow b)].$$

```

In[*]:= RiemannCD[la, lb, lc, ud] // RiemannToGamma

Out[*]=

$$-\partial_a \Gamma_{bc}^d + \partial_b \Gamma_{ac}^d - \Gamma_{ae}^d \Gamma_{bc}^e + \Gamma_{ac}^e \Gamma_{be}^d$$


```

곡률 텐서의 대칭:

$$R_{abcd} = R_{[ab][cd]} = R_{cdab}, \quad R_{[abc]}^d = 0$$

```

In[*]:= GetSymmetry[RiemannCD]
% // GStoString
% // AllPermutations

Out[*]=
GenSet[{Cycles[{{1, 2}}], -1},
{Cycles[{{3, 4}}], -1}, {Cycles[{{1, 3}}, {2, 4}}], 1}]

Out[*]=
-bacd-abdc+cdab

Out[*]=
+abcd-abdc-bacd+badc+cdab-cdba-dcab+dcba

```

Bianchi 항등식:

$$\nabla_{[a} R_{bc]d}^e = 0$$

```
In[*]:= exprBianchi = CD[la, RiemannCD[lb, lc, ld, ue]] +  
          CD[lb, RiemannCD[lc, la, ld, ue]] + CD[lc, RiemannCD[la, lb, ld, ue]]
```

```
Out[*]=
```

$$\nabla_a R_{bcd}^e + \nabla_b R_{cad}^e + \nabla_c R_{abd}^e$$

```
In[*]:= % // CDtoBD  
% // RiemannToGamma  
% // Tsimplify
```

```
Out[*]=
```

$$\partial_a R_{bcd}^e + \partial_b R_{cad}^e + \partial_c R_{abd}^e + \Gamma_{cf}^e R_{abd}^f - \Gamma_{cd}^f R_{abf}^e - \Gamma_{cb}^f R_{afd}^e + \Gamma_{af}^e R_{bcd}^f - \Gamma_{ad}^f R_{bcf}^e -$$

$$\Gamma_{ac}^f R_{bfd}^e + \Gamma_{bf}^e R_{cad}^f - \Gamma_{bd}^f R_{caf}^e - \Gamma_{ba}^f R_{cfd}^e - \Gamma_{bc}^f R_{fad}^e - \Gamma_{ca}^f R_{fbd}^e - \Gamma_{ab}^f R_{fcd}^e$$

```
Out[*]=
```

$$- \partial_a \partial_b \Gamma_{cd}^e + \partial_a \partial_c \Gamma_{bd}^e + \partial_b \partial_a \Gamma_{cd}^e - \partial_b \partial_c \Gamma_{ad}^e - \partial_c \partial_a \Gamma_{bd}^e + \partial_c \partial_b \Gamma_{ad}^e - \partial_c \Gamma_{fd}^e \Gamma_{ab}^f + \partial_f \Gamma_{cd}^e \Gamma_{ab}^f +$$

$$\partial_b \Gamma_{fd}^e \Gamma_{ac}^f - \partial_f \Gamma_{bd}^e \Gamma_{ac}^f + \partial_c \Gamma_{fd}^e \Gamma_{ba}^f - \partial_f \Gamma_{cd}^e \Gamma_{ba}^f - \partial_a \Gamma_{fd}^e \Gamma_{bc}^f + \partial_f \Gamma_{ad}^e \Gamma_{bc}^f -$$

$$\partial_b \Gamma_{fd}^e \Gamma_{ca}^f + \partial_f \Gamma_{bd}^e \Gamma_{ca}^f + \partial_a \Gamma_{fd}^e \Gamma_{cb}^f - \partial_f \Gamma_{ad}^e \Gamma_{cb}^f + \Gamma_{ac}^f \Gamma_{bg}^e \Gamma_{fd}^g - \Gamma_{ab}^f \Gamma_{cg}^e \Gamma_{fd}^g +$$

$$\Gamma_{ba}^f \Gamma_{cg}^e \Gamma_{fd}^g - \Gamma_{ac}^f \Gamma_{bd}^g \Gamma_{fg}^e + \Gamma_{ab}^f \Gamma_{cd}^g \Gamma_{fg}^e - \Gamma_{ba}^f \Gamma_{cd}^g \Gamma_{fg}^e - \Gamma_{af}^e \Gamma_{bc}^g \Gamma_{gd}^f -$$

$$\Gamma_{bf}^e \Gamma_{ca}^g \Gamma_{gd}^f + \Gamma_{af}^e \Gamma_{cb}^g \Gamma_{gd}^f + \Gamma_{ad}^f \Gamma_{bc}^g \Gamma_{gf}^e + \Gamma_{bd}^f \Gamma_{ca}^g \Gamma_{gf}^e - \Gamma_{ad}^f \Gamma_{cb}^g \Gamma_{gf}^e$$

```
Out[*]=
```

$$0$$

따라서 Bianchi 항등식 $\nabla_{[a} R_{bc]d}^e = 0$ 이 증명되었다.

2.7.1 The Ricci and Einstein

리치 텐서, 스칼라 곡률, 아인슈타인 텐서:

$$R_{ab} = R_{apb}^p, \quad R = R_p^p, \quad G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R$$

축약된 Bianchi 항등식과 아인슈타인 방정식:

$$\nabla^a G_{ab} = 0, \quad G_{ab} = 8\pi G T_{ab}$$

Weyl 텐서:

$$C_{abcd} = R_{abcd} - \frac{1}{n-2} [g_{ac} R_{bd} - g_{ad} R_{bc} - (a \leftrightarrow b)] + \frac{1}{(n-1)(n-2)} (g_{ac} g_{bd} - g_{ad} g_{bc}) R$$

공변 도함수의 교환 공식:

$$(\nabla_a \nabla_b - \nabla_b \nabla_a) T_c^d = R_{abc}^p T_p^d - R_{abp}^d T_c^p$$

```
In[*]:= CD[la, lb, T[lc, ud]]  
CommuteCD[{la, lb}, %] (* torsion-free *)
```

```
Out[*]=
```

$$\nabla_a \nabla_b T_c^d$$

```
Out[*]=
```

$$\nabla_b \nabla_a T_c^d - R_{abe}^d T_c^e + R_{abc}^e T_e^d$$